

A Deep Learning Framework for Bone Fracture Detection and Segmentation

DA499 Final Project | Yarmouk University

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Academic Guidance

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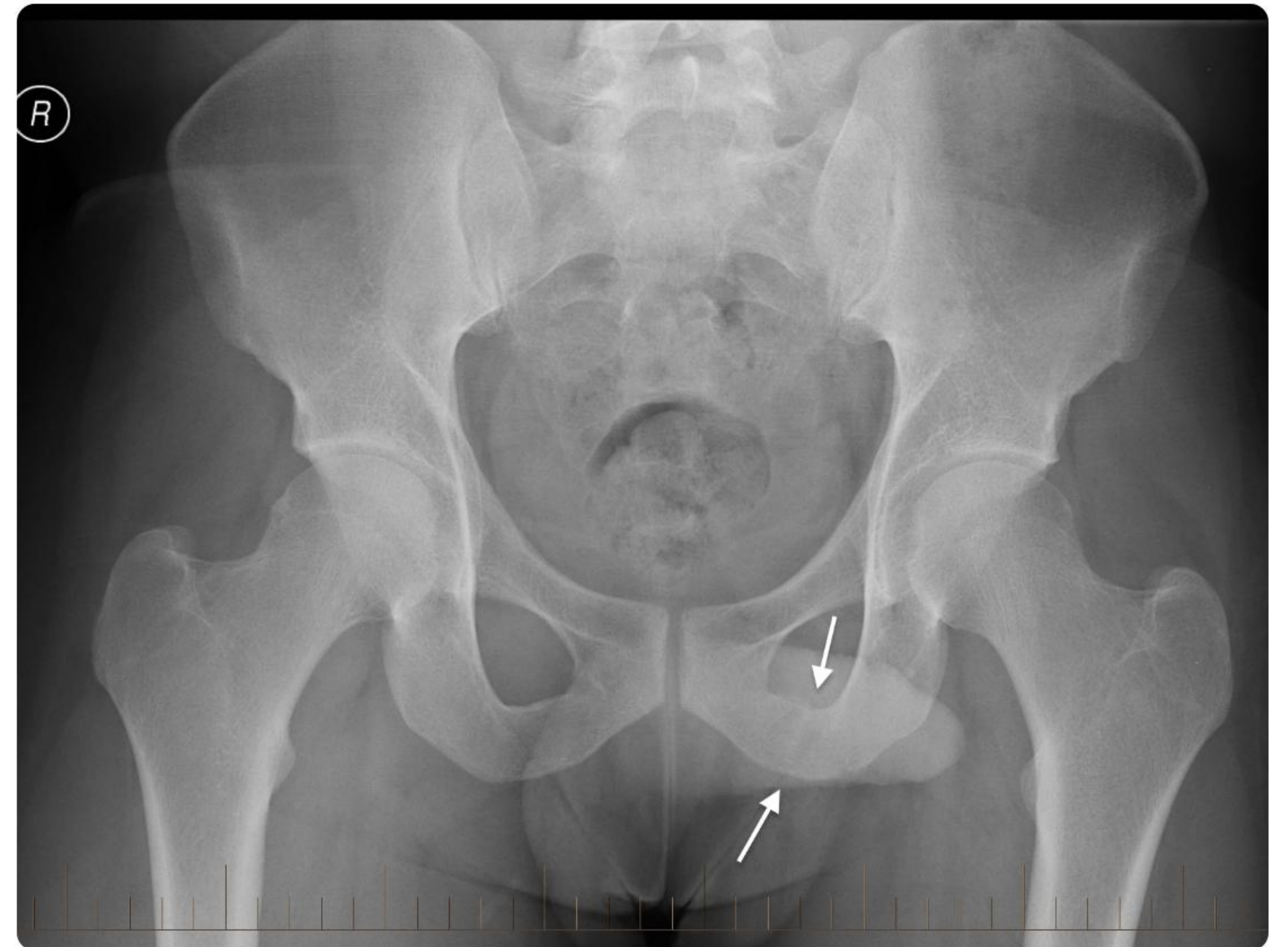
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Introduction

Bone fractures pose significant healthcare challenges. Traditional reliance on manual X-ray interpretation is prone to human error, especially in emergency departments where workload is high.

AI transforms diagnostics by detecting subtle fractures, improving procedural efficiency, and reducing diagnostic errors.



Problem Statement



Diagnostic Challenges

Human error in emergency settings often leads to missing micro or hairline fractures.

This project integrates three tasks: **Classification** (injury presence), **Localization** (Bounding Boxes), and **Segmentation** (pixel-precision boundaries) into one robust framework.

Project Objectives



Multi-Task Model

Develop a model performing classification, location, and segmentation in one pass.



Accuracy

Leverage DenseNet121 for capturing fine radiographic details.



Fairness

Address data imbalance using custom Class Weights.

Literature Review

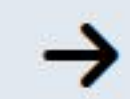
Method	Focus
Clinical Exam	Symptom assessment and mechanism of injury.
X-rays / CT	Gold standards for structural fracture detection.
ML Systems	Tools like BoneView for AI-assisted radiology.

Methodology Workflow

1. Collection



2. Preprocessing



3. Augmentation



4. Training

Preprocessing: Resizing to 224x224, stratified splitting (70/15/15), and utilizing Albumentations for augmentation to ensure generalizability.

Data Analysis (Imbalance)

Non-
Fractured

82.4%

Fractured

17.6%

Strategy: Used class weights and data augmentation to mitigate significant class imbalance.

Results: Performance

Accuracy

Model achieved 90% validation accuracy and 0.92 AUC, indicating robust differentiation capabilities.

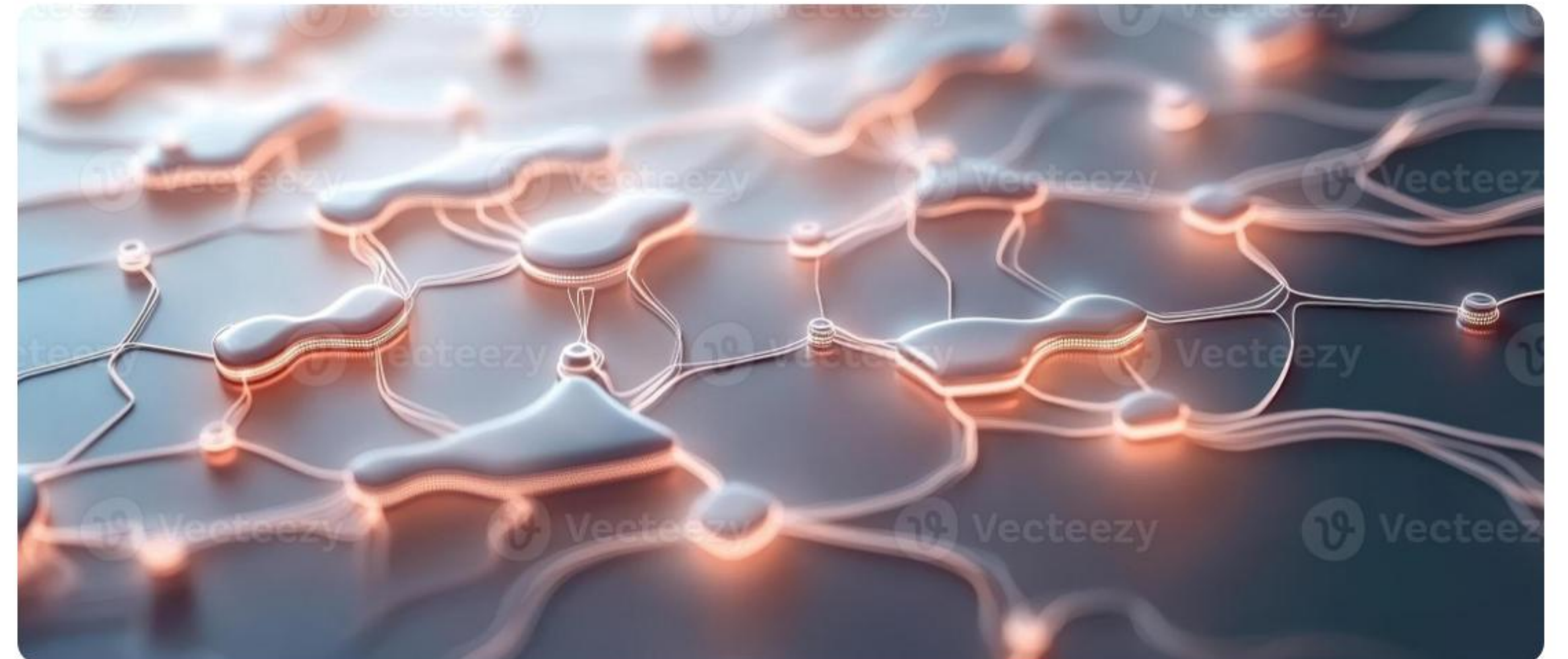
Segmentation

Stable IoU and high accuracy metrics demonstrated consistent identification of fracture boundaries.

Architecture: DenseNet121

The Challenge: Hairline fractures are only a few pixels wide; standard models lose this detail during pooling.

The Solution: DenseNet121 allows dense link information flow, preserving fine details from early layers.



| Limitations

- ⚠ Model bias toward "non-fractured" class due to dataset distribution.
- ⚠ False negatives remain a critical concern in medical deployment.
- ⚠ Dependency on high-quality X-ray input parameters.

Conclusion

The developed DenseNet121 framework demonstrates excellent clinical potential for automated fracture detection. Future work will focus on integrating larger clinical datasets and deploying the model for real-time ER triage support.

Thank You

Questions & Discussion

Image Sources



https://cdn.prod.website-files.com/5f6abdd64a0fa8ba84b72c83/69cc65759ae0f3f25ae58c0f_Pubic%20-%20IMAGE%20ONE.jpg

Source: www.axissportsmedicine.co.nz



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